

IEEE Guide for Interpretation of Gases Generated in Natural Ester and Synthetic Ester-Immersed Transformers

IEEE Power and Energy Society

Sponsored by the
Transformers Committee

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Transformers Committee
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IEEE-SA Standards Board

Abstract: Natural and synthetic ester-immersed transformers are handled in this guide. The following is addressed in this guide:

- a) The theory of combustible gas generation in a natural and synthetic ester-filled transformer
- b) Interpretation of the dissolved gas analysis results
- c) Recommended actions based on the interpretation of dissolved gas analysis results
- d) A bibliography of related literature

Keywords: DGA, dielectric liquid, dissolved gas analysis, high oleic sunflower liquid, IEEE C57.155™, insulating liquid, liquid-immersed transformer, natural ester, rapeseed liquid, soybean liquid, synthetic ester

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1. Overview

1.1 Scope

The guide’s application is for natural and synthetic ester-immersed transformers. This guide addresses the following:

- The theory of combustible gas generation in a natural and synthetic ester-filled transformer.
- Interpretation of the dissolved gas analysis results.
- Recommended actions based on the interpretation of dissolved gas analysis results.
- A bibliography of related literature.

1.2 Purpose

The purpose of this guide is to assist the transformer operator in evaluating dissolved gas analysis (DGA) data obtained from natural ester and synthetic ester-filled transformers.

1.3 Limitation

This guide is based on a limited amount of in-service DGA results and several laboratory-based experimental data sets. Experimental laboratory data can give insight into the insulating liquid's behavior, but it does not replicate actual in-service conditions and fault conditions found in an operating transformer.

Suitably trained personnel following the ASTM D923¹ procedure should perform DGA sampling. The measurement of gas concentrations should be performed by an analytical laboratory according to the methods defined in ASTM D3612, or by suitably trained personnel using a portable instrument, or by means of an online monitoring device. Knowledge of the equipment design, liquid, and solid insulation materials with an understanding of this guide are recommended to perform an adequate interpretation of DGA data. Computer assistance including utilization of spreadsheets, databases, analytical software, maintenance management systems, etc. can be useful because of the large amount of data. The nature of the interpretation also depends upon the context or application of the equipment.

Repair decisions should be based on a variety of tests and inspections that may include multiple confirmation DGA samples. Using a single DGA sample should be avoided as the basis for a repair decision. Results should be confirmed by additional DGA and other tests with expert consultation before performing inspection and/or repairs. Local operating conditions, requirements, and safety issues should be undertaken for the sample.

Many techniques for the detection and the measurement of gases have been established through experience and research. However, it must be recognized that analysis of these gases and interpretation of their significance is at this time not an exact science as there are many variables involved. The presence and quantity of dissolved gases are dependent on many variables such as type of equipment, location, and temperature of the fault; solubility and degree of saturation of various gases in the insulating liquid; the type of liquid preservation system; the type and rate of liquid circulation; the kinds of material in contact with the fault; and finally, variables associated with the sampling and measuring procedures themselves. While oxygen and nitrogen gases are not generated in the insulating liquid, these gases are often used to diagnose issues associated with tank integrity and the sample process.

2. Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document including any amendments or corrigenda applies.

ASTM D923, Standard Practices for Sampling Electrical Insulating Liquids.²

ASTM D3612, Standard Test Method for Analysis of Gases Dissolved in Electrical Insulating Oil by Gas Chromatography.

ASTM D7150, Standard Test Method for the Determination of Gassing Characteristics of Insulating Liquids Under Thermal Stress.

¹ Information on references can be found in Clause 2.

² ASTM publications are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959, USA (<http://www.astm.org/>).

3. Definitions, acronyms, and abbreviations

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. *The IEEE Standards Dictionary: Glossary of Terms & Definitions* should be referenced for terms not defined in this clause.³

dissolved gas analysis (DGA): The extraction, detection, and quantification of gases dissolved in insulating liquid.

high oleic based natural ester insulating liquid: An insulating liquid comprised of a natural ester liquid having a high (greater than 75%) oleic acid (monounsaturated fatty acid, C 18:1) level composition. These liquids can be derived from high oleic sunflower or safflower oils.

high oleic sunflower based natural ester insulating liquid: Natural ester insulating liquid derived from high oleic liquids with a typical fatty acid composition: 82% of oleic acid, 9% of linoleic acid, <1% of linolenic acid.

key gases: Gases generated in insulating liquid-immersed transformers that can be used for qualitative determination of fault types, based on which gases are typical or predominant at various temperatures.

natural ester insulating liquid: Natural ester fluids are derived from an animal or agricultural (vegetable or seed) source.

oxidation inhibitor additives: Any substance added to an insulating liquid to improve its resistance to deleterious attack in an oxidizing environment.

rapeseed-based natural ester insulating liquid: Natural ester insulating liquid derived from rapeseed oil extract, with a typical fatty acid composition: 62% of oleic acid, 22% of linoleic acid, 10% of linolenic acid.

soybean-based natural ester insulating liquid: Natural ester dielectric liquid derived from soybean seed oil extract, with typical fatty acid composition: 24% of oleic acid, 54% of linoleic acid, 7% of linolenic acid.

synthetic ester insulating liquid: An insulating liquid that is a reaction product from the combination of an organic acid and an alcohol such as pentaerythritol.

total dissolved combustible gas: Summation of hydrogen, methane, ethane, ethylene, acetylene, and carbon monoxide gas concentrations.

3.2 Acronyms and abbreviations

C.I.	confidence interval
DGA	dissolved gas analysis
GCMS	gas chromatography mass spectrometry
TDCG	total dissolved combustible gas

³ *The IEEE Standards Dictionary: Glossary of Terms & Definitions* is available at <http://shop.ieee.org/>.

4. General theory

4.1 Background⁴

Synthetic ester insulating liquids have been primarily used as a less flammable liquid substitute in specialty transformer applications since the early 1980s. Natural ester insulating liquids were first used with small distribution transformers in the late 1990s and were introduced for use in power transformers in 2001. With an increasing population of ester-immersed transformers, there is an increasing need for a specific guide to the dissolved gas analysis (DGA) of these liquids.

It is best when a large number of field observations can be used as the basis for defining a normally operating transformer. As yet, there have not been enough failures of ester-immersed transformers to clearly define characteristic transformer faults or risk of failure, so laboratory data is used to provide indication of normal operation, failure conditions, and trending along with a comparison to mineral oils.

The value of DGA derives from the fact that certain gases are produced by the abnormal release of thermal and/or electrical energy into liquid-immersed equipment. Analysis of the dissolved gas is used to indicate the type of fault. Determination of the rates of gas production is used to indicate the severity of the fault.

The differences between an ester liquid and mineral oil are far greater than the differences between any two mineral oils. Fortunately, the gases generated in ester liquids under fault conditions are the same as those that are generated in mineral oil. However, the ratios and rates of generation can be very different. This at least allows for the use of common methods for sampling and testing.

The differences in the chemical structure of ester liquids leads to important differences in how some gases are produced. Gases produced by these different processes give new or additional meaning to the possible interpretation of DGA results. This means that a common method for interpretation of mineral oils data will not work as reliably for ester liquids. Modifications to interpretative methods for mineral oils are necessary for their use with ester fluids.

The most significant differences in gas production for ester fluids are as follows:

- Ethane produced from non-fault conditions for ester fluids containing linolenic acid
- Methane, ethane, and ethylene produced in greater amounts and at lower temperatures from overheating
- Methane, ethane, and ethylene produced in different proportions than mineral oil from overheating
- Carbon dioxide and carbon monoxide produced abundantly from overheating of the ester liquid causes of gas formation

As with mineral oil insulating liquids, the fundamental steps of the decomposition of esters involves the breaking of carbon–hydrogen and carbon–carbon bonds. Active hydrogen and hydrocarbon fragments, called “free radicals” are formed. These free radicals can combine with each other to form gases, molecular hydrogen (H_2), methane (CH_4), ethane (C_2H_6), etc., or they can recombine to form new, condensable molecules. Further decomposition and rearrangement processes lead to the formation of products such as ethylene and acetylene and, in the extreme, to modestly hydrogenated carbon in particulate form.

The two principal causes of gas formation within an operating transformer are thermal and electrical stresses. Heating due to conditions such as overload, excessive conductor resistance, and magnetic flux leakage can produce gases from thermal decomposition of the liquid and the solid insulation. Gases are also produced from the decomposition of liquid and solid insulation exposed to electric discharge. Generally, where decomposition gases are formed principally by low energy and partial discharges, there is little or no

⁴ Reprinted with permission from TJIH2b Analytical Services, Inc., *Understanding Dissolved Gas Analysis of Ester fluids-Part 2 Thermal Decomposition of Ester Fluids*, ©2012.

associated heat. Progressively higher energy discharges will also produce progressively greater amounts of heat energy.

The abundance of the ester groups and carbon chains of sixteen to eighteen carbons combined with the lack of the naphthenic and aromatic structures found in mineral oils give ester liquids a notably different composition. Although many of the gases generated under thermal and electrical stress are the same for mineral oils and ester insulating liquids, there are differences in the production, proportions, and solubility of these gases.

4.1.1 Stray gassing

Stray gassing refers to the gases produced from the liquid under normal operating and overload conditions in the absence of a fault. The laboratory test for measuring stray gas formation, ASTM D7150, is conducted at 120 °C. Characteristic stray gassing occurs in natural esters that contain linolenic acid. Linolenic acid produces ethane as a by-product of oxidation. Soybean and rapeseed-based liquids contain higher amounts of this acid and routinely have measurable amounts of ethane produced under normal operating conditions. Exposure to oxygen, light, and heat affect the amounts of ethane observed, so special consideration should be given to the storage and handling of both bulk fluids and laboratory samples.

The production of ethane from stray gassing can be significant but it has been observed to stabilize or level off over time. The ethane gas concentration can reach several hundred ppm without observing significant increases in other combustible gases as noted in Annex H, [B16], and [B9]. The stray gassing phenomenon may require additional monitoring until the gas stabilizes to establish a trending baseline.

4.1.2 Electrical stress

The relative proportions or ratios of fault gases produced by partial discharge and D1 region from the Duval Triangle Analysis method illustrated in Annex G, in ester fluids are very similar to those produced in mineral oils. Under identical conditions the amount of the discharge fault gases measured is similar for mineral oil and synthetic ester liquids. Under the same conditions the amount of discharge fault gases produced in natural ester liquids is an order of magnitude lower.

To clarify these relationships, if both the liquid volumes and the fault conditions were identical, the concentration (in ppm) of a fault gas would be similar for mineral oil and synthetic ester liquid while the concentration in the natural ester liquid would be approximately ten times lower.

4.1.3 Thermal stress

Ester liquids subjected to sufficient heating produce gases from the thermal decomposition of the hydrocarbon portions of the esters, from the thermal decomposition of the ester groups, and from the dissociated acid groups as well as from stray gassing reactions. The total amount of gas produced tends to be two or more times greater than what is produced from mineral oils as seen in Annex D.

4.1.3.1 Acid production

One of the principal ester liquid pyrolysis reactions is the decomposition of the triglyceride to two free fatty acids, propenal, and the ketene from the third fatty acid. While this reaction produces no gaseous products, each of the products reacts further to produce gases described in the following subclauses. The total acid number rises from this reaction and provides corroboration that the following gases arose from pyrolysis of the ester fluid.

4.1.3.2 Carbon oxides production

As the source temperatures increase, thermal decomposition of ester liquids shows large increases in carbon oxides that correspond to increases in acid content. By comparison, the carbon oxide levels from mineral oil subjected to the same conditions are orders of magnitude smaller. The particular chemistry of ester fluids allows carbon monoxide to be produced in greater amounts than carbon dioxide especially at higher temperatures.

Carbon oxides produced from the ester liquid could greatly exceed the carbon oxides produced from cellulose and confuse interpretations of cellulose involvement. The corresponding use of acid number and ethylene concentration may help to resolve some interpretations of the source of these gases.

4.1.3.3 Ethylene production

Ethylene production in ester liquids becomes substantial at temperatures that are 50 °C to 100 °C lower than for mineral oil and remains substantially greater through the range of higher temperatures. This may arise in part because of reactions that produce ethylene together with carbon monoxide.

Ethylene that is produced by natural ester specific reactions will not exceed the amount of carbon monoxide produced. Yet, at the highest temperatures studied, ethylene production does exceed carbon monoxide production significantly. Coincidentally, this happens in the same temperature range that mineral oil develops substantial production of ethylene. This suggests that below these higher temperatures primarily the natural ester process is occurring, while at these higher temperatures, a process common to mineral oil becomes dominant.

4.1.3.4 Methane and ethane production

As with ethylene, methane and ethane production in ester liquids becomes substantial at temperatures that are 50 °C to 100 °C lower than for mineral oil. At the highest temperatures studied methane and ethane production becomes more similar to mineral oil. Methane and ethane are produced by fracturing the hydrocarbon chains of ester fluids. Coincidentally, through part of the higher temperature range studied methane and ethane are produced in similar quantities. In mineral oils, this is not the case. The differences in the amounts and the proportions of methane and ethane are fundamentally due to the different hydrocarbon structures of ester fluids compared to that of mineral oils.

5. Interpretation of gas analysis

5.1 Cellulose thermal decomposition

Overheating of cellulosic insulation in an ester liquid-immersed transformer will result in the production of both carbon monoxide and carbon dioxide. The ratio of CO₂/CO is temperature dependent and it is typically similar to mineral oil insulating liquid. Hydrocarbon gases are not usually generated in significant quantities from the cellulosic insulation alone. Therefore, the absence of these gases may help distinguish between overheating of the ester liquid and overheating of the cellulosic insulation at temperatures below 400 °C–450 °C. Above this temperature range the production of carbon oxides from the ester liquid will likely mask that produced from cellulose.

Because the paper begins to degrade at lower temperatures than the fluid, its gaseous by-products are found at normal operating temperatures in the transformer. Generator step-up (GSU) transformers for example, typically operate at or near the nameplate rating and they will normally generate several hundred µL/L (ppm) of CO and several thousand µL/L (ppm) of CO₂ without excessive hot spots. The ratio of CO₂/CO is sometimes used as an indicator of the thermal decomposition of cellulose. This ratio is normally more than seven. For the CO₂/CO ratio, the respective values of CO₂ and CO should exceed 5,000 µL/L (ppm) and 500 µL/L (ppm) in order to improve the certainty factor because the ratios are sensitive to minimum values

(IEEE Std C57.104™ [B15]).⁵ As the magnitude of CO increases, the ratio of CO₂/CO decreases. This may indicate an abnormality that is degrading cellulosic insulation.

A large increase in the carbon monoxide and carbon dioxide rates of changes should be investigated as possible cellulose overheating or high temperature overheating of the ester insulating liquid.

5.2 Pyrolysis

Natural and synthetic ester insulating liquids produce combustible gases as shown in the laboratory experimental data in Annex D and Annex E. Pyrolysis near cellulose material may also produce carbon oxide gases.

5.3 Electric discharge decomposition

5.3.1 Partial discharge (corona) decomposition

Hydrogen is the predominant gas generated by partial discharges in an ester-immersed transformer.

5.3.2 Electrical arcing decomposition

The major gases generated during arcing in an ester liquid-immersed transformer are hydrogen, methane, ethane, ethylene, acetylene, carbon monoxide, and carbon dioxide. Hydrogen and the hydrocarbon gases are produced in similar proportions but at different rates compared to mineral oil. The carbon oxides are produced in different proportions and at different rates compared to mineral oil. For more information, see Annex D.

6. Application of DGA to electrical equipment

All transformers generate gases to some extent at normal operating temperatures. But occasionally generated gases indicate an abnormality within an operating transformer such as a local or general overheating, dielectric material issues, or a combination of these items. In electrical equipment, these abnormalities are called “faults.” The behavior of insulating liquids for thermal, partial discharge, and arcing faults are described in Clause 4. Internal faults in the liquid produce the gaseous by-products hydrogen (H₂), methane (CH₄), acetylene (C₂H₂), ethylene (C₂H₄), and ethane (C₂H₆) and recent laboratory experimental data includes the production of carbon monoxide (CO) and carbon dioxide (CO₂). When cellulose is involved, the faults produce methane (CH₄), hydrogen (H₂), carbon monoxide (CO), and carbon dioxide (CO₂). Each of these types of faults produces certain gases that are generally combustible. The analysis of combustible gases may indicate the existence of any one, or a combination, of thermal, electrical, or partial discharge faults. Combinations for each of the separate gases determined by chromatography are unique for different fault temperatures. Also, the ratios of certain key gases have been found to suggest fault types. Interpretation by the individual gases can become difficult when there is more than one fault, or when one type of fault progresses to another type, such as an electrical problem developing from a thermal problem. Dissolved gas sample results are typically compared and trended with historic results, which can help identify a change in transformer condition.

6.1 Threshold transformer gas concentrations

In lieu of the historic sample data from a new transformer or in-service transformers with no previous sample history, DGA sample data can be compared to locally produced threshold gas concentrations or use the percentile data presented in Table 1 for an indication of normality. The statistical analysis does not take

⁵The numbers in brackets correspond to those of the bibliography in Annex I.

the place of diagnostic methods like trending and ratio analysis. The threshold gas concentrations can be used for initial samples from new transformer samples or service aged samples with no previous sample historic for result interpretation to establish a sample frequency. An initial sample that exceeds the threshold concentrations or the values given in Table 1 should have a confirmation DGA sample. Subsequent results of samples should be trended and, if necessary, diagnostic methods applied to the results.

A group of laboratories, analytic service providers, and equipment owners provided in-service transformer dissolved gas results to the IEEE for use in this guideline. The DGA database is maintained by the Insulating Fluid Subcommittee of the IEEE Transformers Committee. A statistical analysis of the contributed DGA database produced the 90th percentile gas concentrations shown in Table 1. Below each 90th percentile value is an estimated 95% confidence interval based on the number of available result records, indicating how much it would be likely to vary if re-derived from a similar database. It is evident that percentiles derived from smaller numbers of sample records (such as for synthetic ester liquids) are much more uncertain than those derived from larger numbers of sample records (such as for soybean-based ester liquids); therefore, the confidence interval range is wider for the synthetic fluid.

Table 1—Threshold value for transformers with no previous sample history, summary of 90th percentile values [$\mu\text{L/L}$ (ppm)] with 95% confidence interval from Annex A⁶

Ester Fluid Type	Number of Records		H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO
Soybean	4,378	90th Percentile	112	20	232	18	1	161
		95% C.I.	(105–118)	(19–22)	(219–247)	(17–20)	(1–1)	(150–179)
High Oleic Sunflower	476	90th Percentile	35	25	58	16	0	497
		95% C.I.	(24–45)	(18–30)	(36–84)	(12–23)	(0–0)	(314–583)
Synthetic	157	90th Percentile	64	104	124	150	13	1344
		95% C.I.	(52–82)	(49–135)	(105–362)	(79–215)	(0–33)	(937–1526)

The 90th percentile for each key gas is reported as the upper end of the range of presumed typical values, based on the available data. Sample records appearing to belong to after-failure oil samples were excluded from the statistical analysis. More information about the analysis with detailed statistical results is provided in Annex A. Data for 95% and 98% percentiles are also given in Annex A.

6.2 Recognition of a gassing problem—establishing operating priorities

From an operational point of view, it is important to establish the following priorities:

- Consistent sample extraction and handling: Using sample procedures that provide a representative liquid sample is the initial step for DGA. Changes in sampling technique and sample container handling can vary the results and affect the diagnoses.
- Repeatable sample analysis: Detect the generation of any gases that exceed expected concentrations and utilize appropriate guidelines so the possible abnormality may be recognized at the earliest practical time in order to minimize damage.
- Result diagnoses: Evaluate the impact of an abnormality on the serviceability of the transformer, using a set of guidelines or recommendations.

⁶ No available data for rape seed insulating liquid.

- Follow-up of results: Beginning with increased surveillance and confirming or supplementary analysis and leading to either a determination of load sensitivity, operating conditions, or removing the unit from service.

There is a limited amount of information available for field-based natural and synthetic ester-immersed transformers. A significant volume of information has been generated about diagnosing incipient fault conditions in mineral oil-immersed transformers. This information is of a general nature but is often applied to very specific problems or situations. One consistent finding with all schemes for interpreting gas analysis is that the more information available concerning the history of the transformer and test data, the greater the probability for a correct diagnosis of the health of the unit.

A number of simple schemes employing principal gases or programs using ratios of key gases have been employed for providing a tentative diagnosis when previous information is unavailable or indicated no fault condition existed. Principal gas or ratio methods require detectable or minimum levels of gases to be present, or norms to be exceeded, before they can provide a useful diagnosis (See Annex D, Annex E, and Annex G for additional information).

6.3 Suggested operating procedures utilizing the detection and analysis of combustible gases

Samples for establishing baseline values for future samples should be taken prior to energizing the equipment. This sample is used to confirm the liquid's initial condition and the evolution of combustible gases due to equipment operation. Gas sources other than transformer operation can include contamination from used or service-aged insulating liquid, previously processed fluids, and faulty fluid handling equipment like fluid transfer pumps. There have also been cases of residual combustible gases from inadequate purges that followed tank welding process and reactions of incompatible materials. Equipment can be designed with components that perform energized and/or load-break switching operations and internal expulsion fusing. Such devices will produce combustible gases similar to electrical breakdowns so their presence should be determined prior to analysis of results.

DGA samples following energizing and loading can be used to verify equipment operation. An example for new equipment or equipment status change would be DGA samples at 24-hour, 48-hour, and 72-hour time periods to verify proper equipment operation. Subsequent samples would include a monthly, quarterly, and biannual DGA samples until reaching the routine sample frequency. Changes in the DGA gas profile or expected rates of change should be checked with a confirmation DGA sample followed by an investigation into the gassing source.

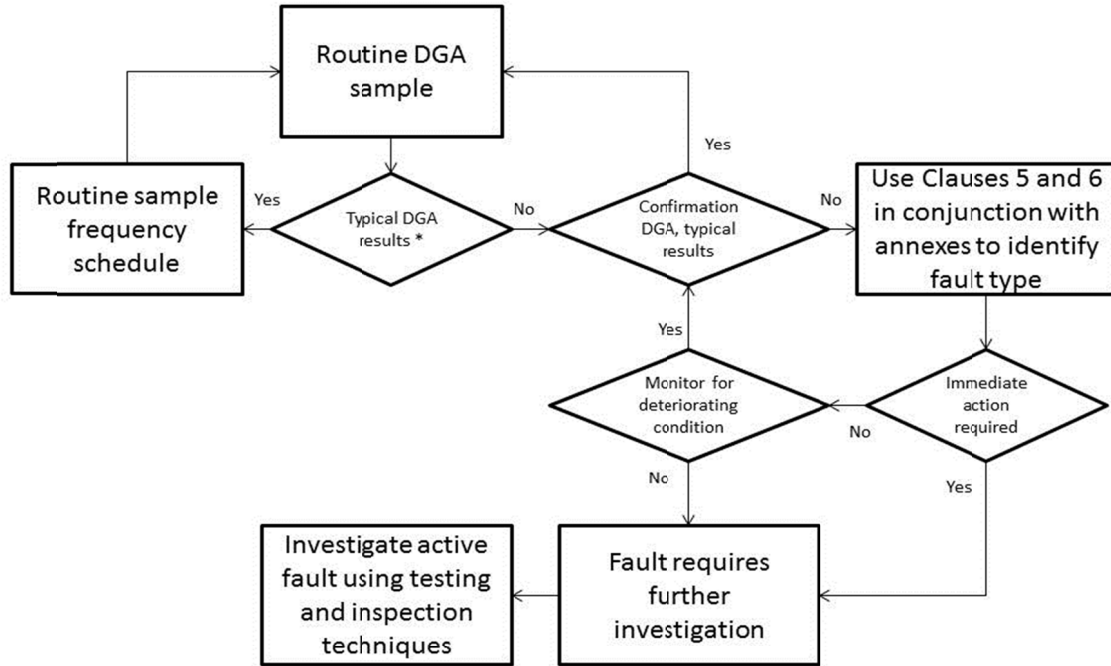
Actions following the identification of an active fault depend on the fault severity, the importance of the transformer application, and potential consequence of an unplanned outage. Other factors include spare equipment availability, cost, environmental and safety risks, and time to replace for any production interruption and equipment removal difficulty. When sudden increases in the dissolved gas content of the insulation liquid in operating transformers occur and an active fault is suspected, an increasing combustible gas generation rate indicates a problem of increasing severity; therefore, a shorter sampling interval is recommended to monitor the problem or an intervening action like electrical testing and/or internal inspection.

Applying the laboratory experiment gas profiles found in the annexes can assist with identifying a possible gassing source. Once the source of gassing is determined by analysis, inspection, consultation, (or combinations thereof), and the risk has been assessed, then engineering judgment should be applied to determine the sampling interval and operating procedure.

The data found in Annex B and Annex C can be used to help identify non-fault-related generated combustible gases formed from near normal operating temperatures or samples exposed to certain environmental factors like sunlight. The laboratory experimental data presented in Annex D can be applied to gas profiles and Duval Triangle analysis method found in Annex G to trend and monitor a fault

condition. Applying these analytic methods to low combustible gas concentrations will provide incorrect diagnostic results.

Figure 1 provides a typical decision flow chart for DGA analysis.



*Compare to transformer historic insulating liquid sample results

Figure 1 —Routine sample flowchart of suggested operating procedure utilizing the detection and analysis of combustible gases in liquid-immersed transformers

6.3.1 Atmospheric gases

Ester liquid-immersed transformers will typically have a sealed or gas blanketed liquid preservation system to protect the insulating liquid from exposure to excessively high oxygen gas concentrations. Commonly used for transformers, nitrogen is an inert gas that is typically reported in DGA results. The nitrogen sources can include initial dissolved nitrogen, nitrogen absorbed from air, or dry nitrogen in the headspace of the transformer or conservator. Lower than expected nitrogen gas concentration, elevated oxygen gas concentration, or elevated oxygen to nitrogen ratio may indicate a poor quality sample or transformer tank integrity issues. Steps should be taken to identify the atmospheric gas source, correct the problem, and assess the amount of fluid deterioration, if any.

6.3.2 Cellulose

Elevated carbon monoxide (CO) and carbon dioxide (CO₂) gas concentrations may indicate higher than expected temperatures near cellulose material. The amounts and rates of gas generation can vary based on paper type, exposure temperature, amount of cellulose exposed to the energy, time of exposure and the condition of the insulating liquid. Carbon monoxide and carbon dioxide are normal by-products of the cellulose aging process so equipment operating age must be taken into account during the evaluation process.

Elevated carbon monoxide and carbon dioxide gas concentrations should prompt a review of the loading history, operation temperature history and cooling system operation. Evaluation of the combustible gases other than carbon monoxide can help identify a possible gassing source like pyrolysis. Accelerated cellulose aging and burning cellulose from arcing activity can produce higher rates of change for carbon monoxide and carbon dioxide gases. The carbon monoxide production is dependent on the local oxygen gas concentration.

6.3.3 Combustible gases

Elevated temperatures in contact with the insulating fluid will produce combustible gases. The gas concentrations and rates of changes depend on the type of insulating liquid, quantity of liquid exposed to the elevated temperature, and the amount of exposure time. Changes in the combustible gas ratios from the various analysis methods or changes in the individual rates of changes can indicate deterioration in the equipment condition.

Significant changes in the types of combustible gases, rates of changes, and ratio analyses should prompt additional actions. A confirmation sample is taken following any required emergency actions. Confirmation samples are used to rule out external error sources and to confirm variations in the DGA trend. Increasing rates of changes indicate a deteriorating condition while stable combustible gas concentrations indicate an inactive gassing source.

6.3.3.1 Arcing activity

Arcing activity can have an immediate effect on the insulation system by producing gas bubbles and combustion by-products that further deteriorate the insulation system.

6.4 Determining average combustible gas generating rates

Recognition and evaluation of ongoing or recent combustible gas production is an important part of transformer fault detection. When a high concentration of any combustible gas is detected by a laboratory-based sample program, it may be impossible to know whether the gas was formed suddenly, intermittently, or gradually over time. To evaluate the nature and severity of the fault, it is necessary to collect one or more additional DGA samples at suitably short intervals to determine whether significant combustible gas production is continuing and, if so, whether it appears to be sporadic, steady, or accelerating and whether the pattern of gases produced appears to be changing. Online DGA monitoring can provide a higher number of samples over a period of time to reduce the measurement uncertainties (CIGRÉ [B6]).

If gas production is reasonably suspected to be gradual, it can be useful to calculate the average rate of change ($\mu\text{L/L}[\text{ppm}]$ per day) between samples for each key gas by dividing the amount of increase by the number of days between the samples. If an average rate of change in liter (L)/day is desired, multiply the $\mu\text{L/L}$ (ppm) per day average rate of change by the volume (L) of the transformer's main tank and then divide by one million. The rates of change over successive similar sampling intervals can be compared to determine whether gas production is speeding up or slowing down. If gas production is sporadic, evaluation of average rates of change may not be useful.

An average gas evolution rate limit based on field experience has not been established to evaluate the severity of an active internal fault.

Care should be taken when comparing gas concentrations because of errors associated with sampling, sample reproducibility, and the repeatability of the measurement method. Gas increments and rates of increase are meaningful only when the amount of increase is substantially greater than that which could be seen by chance, given the repeatability of the gas concentration measurements.

7. Procedures for obtaining samples

All samples of insulating liquid from electrical apparatus being taken for the purpose of DGA should be taken in accordance with ASTM D923 or similar procedure (ASTM D3305 [B1], ASTM D3613 [B2]). Natural ester insulating liquids are susceptible to stray gassing due to interaction with sunlight and ultraviolet light from florescent lights. For additional information, see Annex C. The sample containers must be stored in dark, cool locations and returned to the laboratory for analysis as soon as possible after drawing the sample.

Annex A

(informative)

DGA database statistical analysis (Dukarm [B8])

A.1 Data preparation

DGA data files from multiple sources referred to in 6.1 were processed to extract only the data belonging to transformers filled with ester-based insulating liquid. After preliminary inspection of the data from each source, a script was written to do most of the work, which consisted of tasks such as the following:

- Change transformer serial numbers to anonymous identifiers
- Map differently named variables into fields with a standardized name
- Remove embedded punctuation marks and “garbage” characters from the data
- Re-code class variables such as cooling types, kV classes, fluid type, and others to a common set of names
- Change sample dates to a common format

The data preparation script produced a combined data file, which was manually inspected and edited to produce the master data file for the statistical analysis.

A.2 Statistical analysis

An initial exploratory analysis showed that some very extreme combustible gas concentrations were present in the data, apparently from after-failure samples. After some consideration, it was decided to exclude all samples with $H_2 + CH_4 + C_2H_4 + C_2H_2 > 10,000 \mu\text{L/L}$ (ppm) as presumed after-failure samples. This eliminated approximately one thousandth of all the ester liquid samples.

An R script was prepared to carry out the statistical analysis and display the results in tables that could be copied and pasted into a report. R is an open-source statistical scripting language that is widely used by professional statisticians and scientists. It is available for free for all commonly-used personal computer operating systems from the R Project for Statistical Computing (<http://www.r-project.org/>). The statistical script generated basic descriptive statistics for all the key gases. One set of statistics was based on the entire data set, and another was based on only the latest sample for each transformer. Although restriction to the latest samples seems reasonable as a way to avoid bias, it turned out to make little difference for this analysis, where the database contains only a few samples each from 3,900 transformers. Descriptive statistics were generated for various subsets of the data, namely synthetic esters, high oleic sunflower, and soybean-based natural ester insulating liquids. Candidate DGA limits based on the conventional 90th, 95th, and 98th percentiles were derived for hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), acetylene (C_2H_2), and carbon monoxide (CO). The limits were calculated for all of the fluid type subsets.

Smooth bootstrapping was used to estimate a 95% confidence interval for each of the limits, roughly indicating the degree of variability to be expected when deriving corresponding limits from a similar database. This involved creating a new data set and regenerating the limits ten thousand times by sampling with replacement from the original data set and adding random simulated measurement noise (relative $\pm 10\%$) to each gas concentration.

Table A.1 through Table A.6 report descriptive statistic information and percentiles for natural ester and synthetic insulating liquids based on the previously described analysis method.

Table A.1—Soybean-based natural ester insulating liquid database descriptive statistics for key gases [$\mu\text{L/L}$ (ppm)]

Gas	Number of records	Min	Q1	Median	Q3	Max	IQR	Max_excl
H ₂	4,378	0	8	24	55	4,853	47	26,149
CH ₄	4,378	0	2	5	10	2,126	8	46,056
C ₂ H ₆	4,378	0	3	28	106	2,881	103	30,000
C ₂ H ₄	4,378	0	1	3	7	4,962	6	200,000
C ₂ H ₂	4,378	0	0	0	0	1,638	0	10,346
CO	4,378	0	25	47	82	10,340	57	22,648
CO ₂	4,378	17	265	465	1,039	89,890	774	24,353

Table A.2—Percentile-based limits for soybean-based natural ester insulating liquid key gases with 95% confidence intervals [$\mu\text{L/L}$ (ppm)]

Gas	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO
90th Percentile	112	20	232	18	1	161
95% C.I.	(105–118)	(19–22)	(219–247)	(17–20)	(1–1)	(150–179)
95th Percentile	171	41	389	36	3	462
95% C.I.	(160–188)	(34–49)	(351–444)	(33–44)	(2–3)	(385–562)
98th Percentile	288	111	841	118	10	1,471
95% C.I.	(261–351)	(95–124)	(764–934)	(88–155)	(7–13)	(1,085–1,684)

Table A.3—High oleic sunflower based natural ester insulating liquid database descriptive statistics for key gases [$\mu\text{L/L}$ (ppm)]

Gas	Number of records	Min	Q1	Median	Q3	Max	IQR	Max_excl
H ₂	476	0	0	3	11	1,755	11	23
CH ₄	476	0	0	2	7	7,187	7	5
C ₂ H ₆	476	0	0	1	6	15,150	6	4
C ₂ H ₄	476	0	0	1	4	1,720	4	1
C ₂ H ₂	476	0	0	0	0	2,819	0	0
CO	476	0	14	59	129.2	3,122	115.2	113
CO ₂	476	13	142.8	325	1,567	9,893	1424.2	845

Table A.4— Percentile-based limits for high oleic sunflower based natural ester insulating liquid key gases with 95% confidence interval [$\mu\text{L/L}$ (ppm)]

Gas	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO
90th Percentile	35	25	58	16	0	497
95% C.I.	(24–45)	(18–30)	(36–84)	(12–23)	(0–0)	(314–583)
95th Percentile	60	37	131	40	0	715
95% C.I.	(48–84)	(31–41)	(92–152)	(25–43)	(0–1)	(595–938)
98th Percentile	126	45	175	46	3	1,335
95% C.I.	(78–482)	(40–210)	(147–230)	(43–54)	(1–14)	(894–2,093)

Table A.5—Synthetic ester insulating liquid database descriptive statistics for key gases [μL/L (ppm)]

Gas	Number of Records	Min	Q1	Median	Q3	Max	IQR	Max_excl
H ₂	157	0	3	12	29	1,489	26	2,758
CH ₄	157	0	5	18	32	1,935	27	2,310
C ₂ H ₆	157	0	3	12	86	672	83	445
C ₂ H ₄	157	0	5	11	28	4,059	23	3,086
C ₂ H ₂	157	0	0	0	0	129	0	2,285
CO	157	22	195	339	495	2,537	300	2,019
CO ₂	157	281	2,406	4,717	6,916	12,510	4,510	7,398

Table A.6—Percentile-based limits for synthetic ester insulating liquid key gases with 95% confidence interval [μL/L (ppm)]

	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO
90th Percentile	64	104	124	150	13	1344
95% C.I.	(52–82)	(49–135)	(105–362)	(79–215)	(0–33)	(937–1,526)
95th Percentile	88	144	474	230	42	1,541
95% C.I.	(67–109)	(109–182)	(133–590)	(157–329)	(18–54)	(1,356–1,845)
98th Percentile	100	179	592	325	51	1736
95% C.I.	(85–1,335)	(139–1,686)	(424–674)	(220–3,359)	(36–93)	(1,555–2,393)

A.3 Results of the analysis

The results of the analysis for soybean-based, high oleic sunflower, and synthetic ester insulating liquids are presented in Table A.1 to Table A.6. For each of the three liquid types, there is a table of key gas descriptive statistics and a table of key gas percentile-based limits.

The column names used for the descriptive statistics include N (number of samples included in the calculations), Min (minimum value), Q1 (first quartile, i.e., 25th percentile), Median (50th percentile), Q3 (third quartile, i.e., 75th percentile), Max (maximum value), IQR (interquartile range, i.e., Q3–Q1), and Max_excl (maximum of the values excluded from the calculations). Recall that samples with H₂ + CH₄ + C₂H₄ + C₂H₂ > 10,000 ppm were excluded as presumed after-failure samples.

For the percentile-based limits, the column names are P90, P95, and P98, designating the 90th, 95th, and 98th percentiles, respectively. Each percentile value is accompanied by an estimated 95% confidence interval showing roughly how much that quantity could be expected to vary if derived from a different but similar database. The limit confidence intervals for the soybean-based ester insulating liquids are narrow because of the large number of samples for that kind of fluid. For the synthetic ester liquids, the small number of samples (157) causes these high percentiles to be uncertain, so that the confidence intervals for neighboring percentiles overlap. Some of the P98 limits are extremely uncertain, i.e., P98 limits derived from another similarly small set of samples are likely to be considerably different.

Annex B

(informative)

ASTM D7150 stray gas results (Hanson, et. al [B1])

ASTM D7150 stresses the insulating liquid under controlled conditions. These gases can be mistaken for a fault condition that would prompt possible unnecessary actions. Table B.1 reports the results for several natural and synthetic ester fluids that can be compared with sample results to help identify non-fault related gassing to other issues.

Table B.1—ASTM D7150 modified stray gas using air sparge, 24-hour test [$\mu\text{L/L}$ (ppm)]

Ester Fluid Type	Synthetic Ester Liquid	High Oleic Sunflower Ester Liquid	Soybean Ester Liquid	Rapeseed Ester Liquid
α -linolenic acid content	0	Approx. 0.2%	Approx. 7%	Approx. 9%
H_2	175	357	316	289
CH_4	14	21	10	60
C_2H_6	0	4	563	3,694
C_2H_4	2	8	7	19
C_2H_2	0	0	0	0
CO	123	203	408	503
CO_2	726	876	1,330	1,696

Annex C

(informative)

Stray gassing due to fluorescent light exposure (Hanson, et al. [B10])

The laboratory experiment data is provided as a comparison of ethane production from exposure to light sources. The fluorescent light sample preparation followed the modified ASTM D7150 test method (air sparge prior to fluorescent light preparation, 1 day, 120 °C). The ASTM D7150 method was modified by reducing the amount of time the sample was exposed to the test method to 1 day. The soybean-based insulating liquid contained about 7% of α - linolenic acid.

The laboratory experiment is used to illustrate the need to protect any fluid sample from sunlight and fluorescent light. Exposure to these light sources can generate gases that can be mistaken as being produced by a transformer fault.

Table C.1—Soybean-based ester liquid DGA results [$\mu\text{L/L}$ (ppm)] exposure to fluorescent light⁷

	ASTM D7150	Fluorescent light only	ASTM D7150 and fluorescent light
	Control	Control	Sample
H ₂	149	257	762
CH ₄	35	23	73
C ₂ H ₆	615	642	2,192
C ₂ H ₄	7	6	20
C ₂ H ₂	0	0	0
CO	220	197	474
CO ₂	942	996	1,998

⁷ Experimental data is limited to soybean-based ester insulating liquid.

Annex D

(informative)

Laboratory pyrolysis experiment information

These studies (Hanson, et al. [B10], Hanson [B11]) were conducted in the laboratory using an experimental configuration summarized in the following section. The experiment did not include materials normally found in an operational transformer, so these results do not represent conditions found in field applications. Such equipment has cellulose material and materials that may react with the insulating liquid to produce different gasses and gas concentrations.

The experimental equipment and configuration are presented in the referenced source and are summarized in this annex.

- a) To allow for observation and measurement of gases formed from insulating liquids at temperatures from 250 °C to 700 °C.
- b) To conduct studies in a fixed volume system to measure relative gas pressures.
- c) To conduct studies in an expandable volume system at atmospheric pressure.
- d) To allow for collection and analysis of liquid and gas samples.

The setup includes an expansion chamber that is maintained at atmospheric pressure. This chamber includes a valve for isolation; a pressure gauge; a defined volume gas space that can be sealed by the isolation valve of the expansion chamber; a liquid reservoir; and a pump that circulates liquid through tubing from the reservoir to the oven and back again into the oven.

The data from studies on soybean, high oleic sunflower, rapeseed, synthetic, and mineral oil insulating liquids are shown in Table D.1 to Table D.5. This data represents the total gas generated for each trial. The total is the combination of gas measured from the gas phase and the gas dissolved in the insulating liquid.

A time of eight hours was selected for the trials because it allowed for detection of gases at 250 °C and management of gases at 700 °C. The insulating liquid volume was 1.5 liters. Note that gas volumes in excess of 100 liters were generated for ester liquids at 700 °C.

The experimental data gives an indication on how the insulating liquid reacts under laboratory conditions. It can be inferred that the liquids will generate gases similar to those presented in the experiment when exposed to similar temperatures. Transformer construction, type of gassing source, duration of exposure, amount of liquid exposed, and other conditions can affect the various gases and concentrations so variations can be expected when diagnosing a potential transformer fault.

Table D.1—Laboratory experience for pyrolysis of soybean-based natural ester liquid [$\mu\text{L/L}$ (ppm)]

Soybean fluid	250 °C	300 °C	350 °C	400 °C	450 °C	500 °C	550 °C	600 °C	650 °C	700 °C
Hydrogen	183	242	453	448	14,578	92,064	449,630	951,663	1,898,615	5,336,773
Methane	9	18	36	163	5,427	32,265	426,389	2,555,117	5,176,221	15,591,740
Ethane	321	404	495	896	12,584	46,790	518,002	2,666,376	5,163,518	11,721,018
Ethylene	2	3	9	59	12,620	66,570	549,371	6,377,429	13,599,997	37,426,253
Acetylene	0	0	0	0	0	0	0	3,573	8,213	33,494
Carbon Monoxide	138	250	388	1,267	159,834	846,321	4,476,588	8,535,474	9,782,953	15,715,859
Carbon Dioxide	528	565	660	1,324	150,631	663,328	3,110,067	5,863,817	6,839,063	9,506,162
Acid Number (mg KOH/g)	0.026	0.049	0.084	0.328	9.035	>10	>10	>10	>10	>10

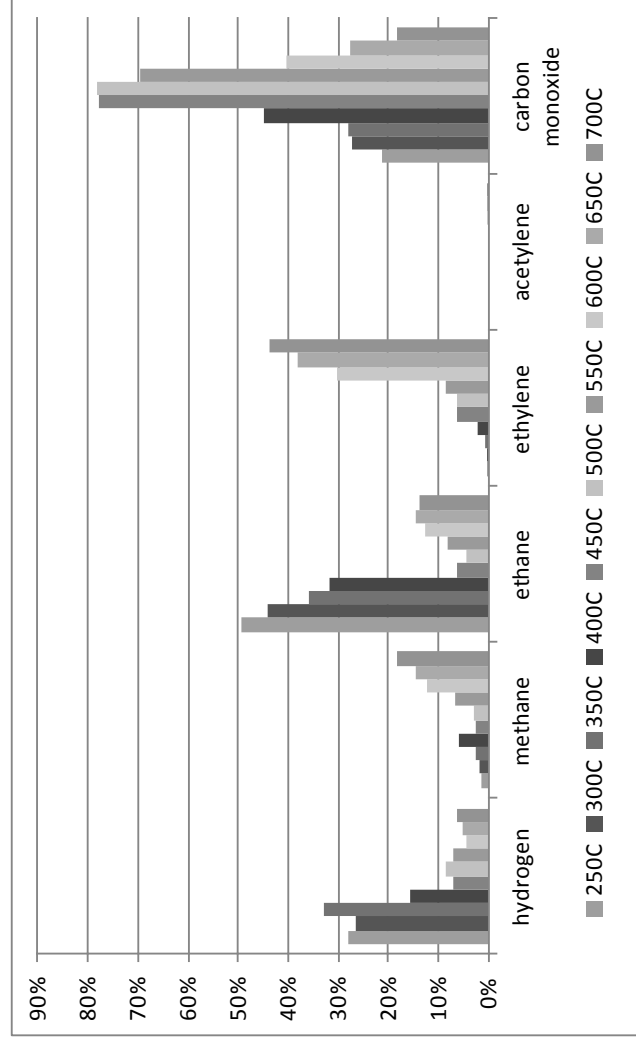


Figure D.1—Pyrolysis of soybean-based ester liquid as % of TDCG

Table D.2—Laboratory experience for pyrolysis of high oleic sunflower ester-based insulating liquid [$\mu\text{L}/\text{L}$ (ppm)]

High oleic sunflower fluid	250 °C	300 °C	350 °C	400 °C	450 °C	500 °C	550 °C	600 °C	650 °C	700 °C
Hydrogen	0	0	0	18	567	23,687	240,787	365,305	1,338,912	3,614,829
Methane	3	9	8	20	1,287	18,503	314,820	1,365,578	6,801,996	16,016,581
Ethane	0	3	3	8	360	19,455	325,411	1,469,350	5,269,037	9,183,573
Ethylene	0	0	0	5	807	45,684	461,567	3,456,086	16,687,658	35,155,966
Acetylene	0	0	0	0	0	0	0	1,580	20,197	52,615
Carbon Monoxide	54	60	77	224	23,192	550,591	4,251,527	4,052,783	7,611,417	12,554,284
Carbon Dioxide	1,003	1,346	1,281	1,822	13,839	489,273	3,195,203	3,181,084	4,581,705	7,989,692
Acid Number (mg KOH/g)	0.043	0.061	0.066	0.15	0.262	8.762	>10	>10	>10	>10

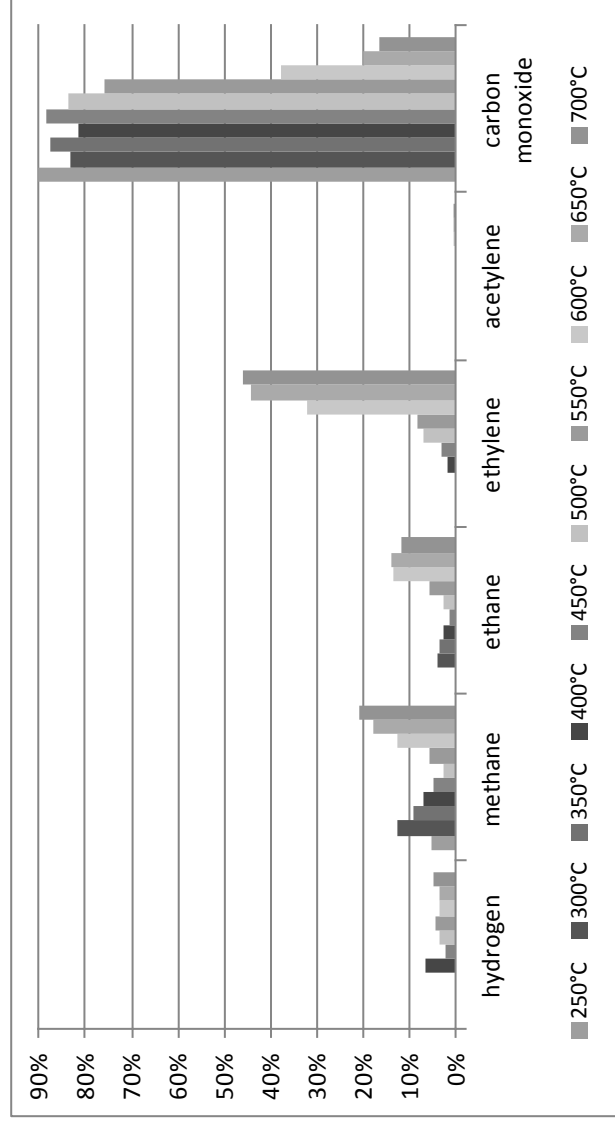


Figure D.2—Pyrolysis of high oleic sunflower-based ester insulating liquid as % of TDCG

Table D.3—Laboratory experience for pyrolysis of rapeseed-based ester insulating liquid [$\mu\text{L/L}$ (ppm)]

Rapeseed fluid	250 °C	300 °C	350 °C	400 °C	450 °C	500 °C	550 °C	600 °C	650 °C	700 °C
Hydrogen	212	295	323	609	10,075	66,135	283,072	775,334	1,697,857	4,206,957
Methane	47	127	113	281	3,987	29,064	368,510	2,650,547	6,970,984	14,354,645
Ethane	3,068	2,930	3,203	4,527	11,401	47,501	451,072	2,848,721	5,891,928	12,718,750
Ethylene	9	54	42	107	5,824	52,620	481,941	6,844,115	17,030,156	34,413,530
Acetylene	0	0	0	0	0	0	0	3,338	35,836	48,392
Carbon Monoxide	589	1,460	1,451	3,352	97,494	679,896	3,716,079	6,317,808	7,906,603	12,585,097
Carbon Dioxide	587	968	995	1,888	63,558	532,926	2,718,221	4,227,107	4,609,595	7,895,977
Acid Number (mg KOH/g)	0.078	0.095	0.124	0.403	5.424	6.839	>10	>10	>10	>10

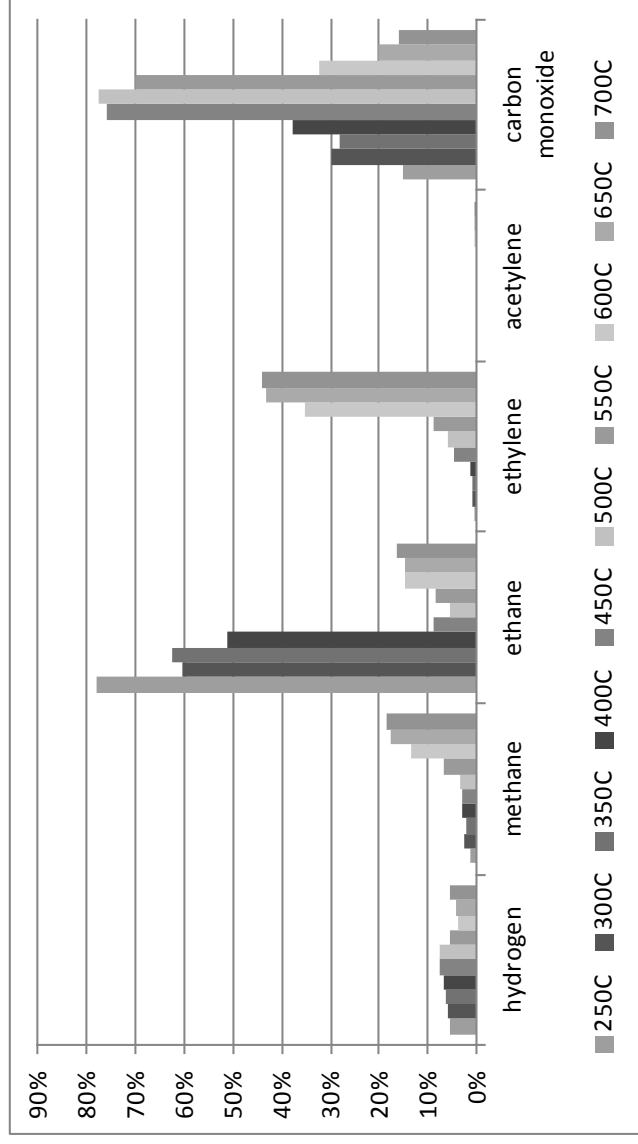


Figure D.3—Pyrolysis of rapeseed-based ester insulating liquid as % of TDCG

Table D.4—Laboratory experience for pyrolysis of synthetic ester insulating liquid [$\mu\text{L/L}$ (ppm)]

Synthetic fluid	250 °C	300 °C	350 °C	400 °C	450 °C	500 °C	550 °C	600 °C	650 °C	700 °C
Hydrogen	2	6	26	20	345	5,848	17,990	108,843	395,394	2,353,482
Methane	2	18	63	146	1,811	24,969	235,443	1,364,812	4,088,106	11,821,529
Ethane	3	36	146	259	2,791	32,202	412,681	1,533,854	3,642,452	7,331,319
Ethylene	6	9	65	168	1,822	21,527	596,456	2,983,722	8,444,719	24,418,135
Acetylene	0	0	0	0	0	0	0	408	2,939	45,931
Carbon Monoxide	63	104	409	617	5,794	101,301	398,553	1,731,122	4,802,909	12,123,174
Carbon Dioxide	524	645	841	946	3,728	62,454	701,822	2,407,159	5,452,250	11,499,587
Acid Number (mg KOH/g)	0.023	0.03	0.054	0.065	0.168	0.288	1.812	4.448	8.912	>10

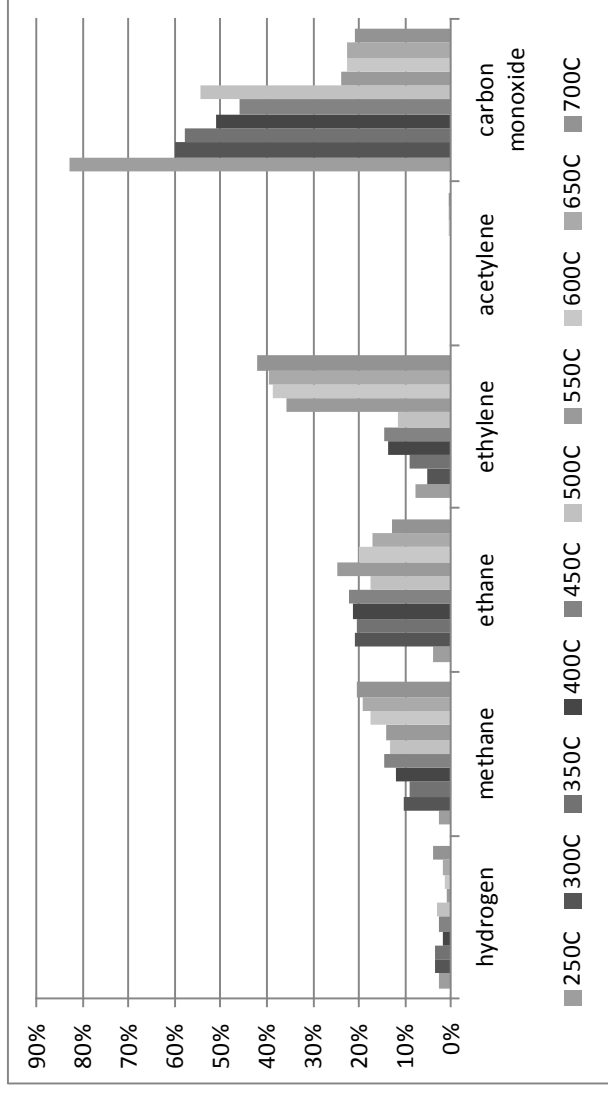


Figure D.4—Pyrolysis of synthetic ester insulating liquid as % of TDCG

Table D.5—Laboratory experience for pyrolysis of mineral oil insulating liquid [$\mu\text{L/L}$ (ppm)]

Mineral oil fluid	250 °C	300 °C	350 °C	400 °C	450 °C	500 °C	550 °C	600 °C	650 °C	700 °C
Hydrogen (H_2)	0	61	65	66	308	635	3,415	112,760	1,349,634	7,248,321
Methane (CH_4)	15	32	77	82	281	1,135	14,060	525,596	5,562,060	18,288,865
Ethane (C_2H_6)	0	9	9	17	181	606	7,688	403,924	2,989,009	7,135,061
Ethylene (C_2H_4)	0	0	2	3	26	502	8,254	528,885	5,156,563	15,505,831
Acetylene (C_2H_2)	0	0	0	0	0	0	0	0	1,506	4,140
Carbon Monoxide (CO)	23	89	156	173	220	256	166	478	5,458	12,096
Carbon Dioxide (CO_2)	392	451	540	553	699	653	866	4,524	51,091	121,840
Acid Number (mg KOH/g)	0.010	0.010	0.010	0.010	0.011	0.010	0.010	0.047	0.048	0.048

The Duval Triangle ratios are calculated for the experimental pyrolysis DGA data presented in this annex. Each ratio is charted in Figure D.5, Figure D.6, and Figure D.7, for the experiment insulating liquids to represent how the ratios change with temperature. The ratios may be useful for evaluating the DGA result for possible fault condition. The ratio calculation and method usage are found in Annex G.

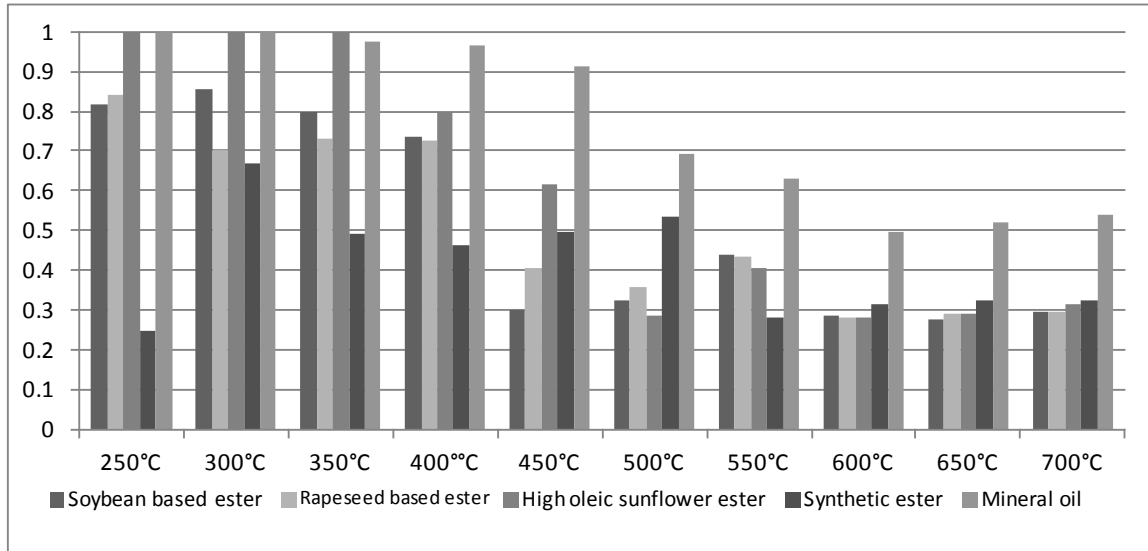


Figure D.5—Duval triangle diagnostic method CH_4 ratio from pyrolysis experiments

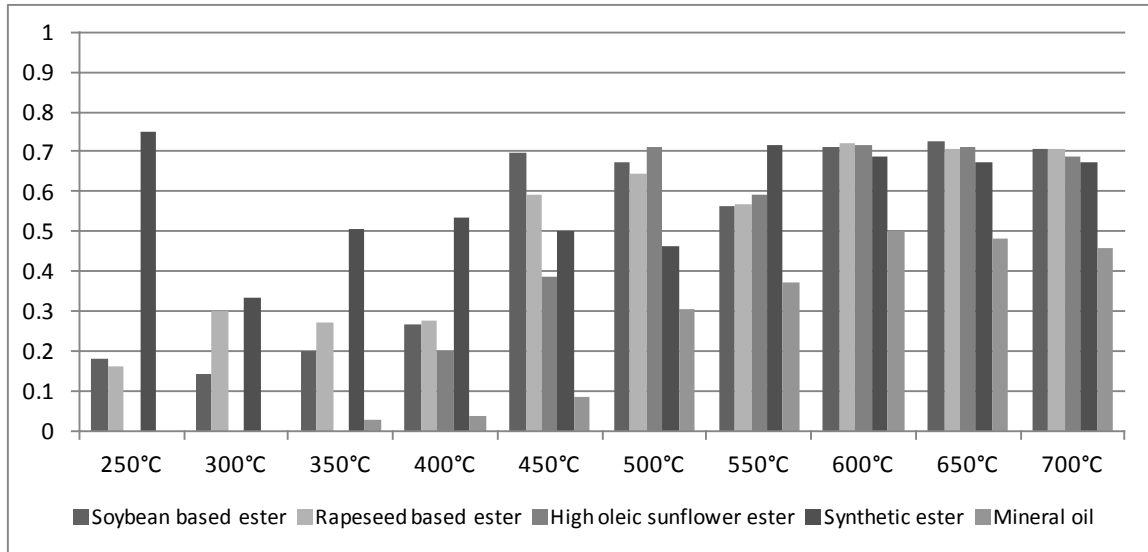


Figure D.6—Duval triangle diagnostic method C_2H_4 ratio from pyrolysis experiments

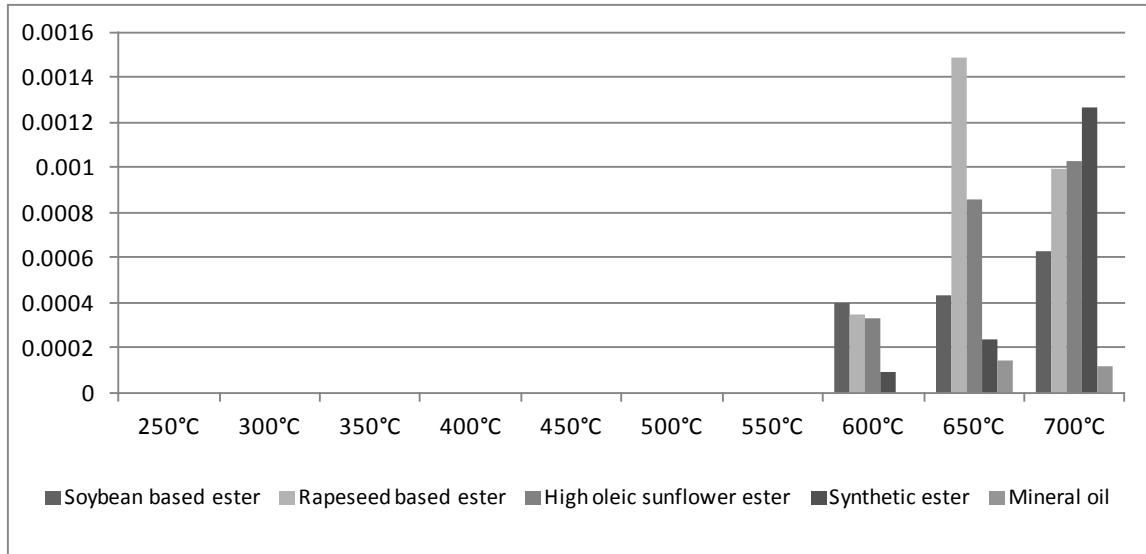


Figure D.7—Duval triangle diagnostic method C_2H_2 ratio from pyrolysis experiments

Annex E

(informative)

EPRI laboratory pyrolysis and arcing experiment information (DGA Interpretation in Natural Ester Fluids [B7])

E.1 Pyrolysis EPRI laboratory experiment summary

This experiment differs from the one outlined in Annex D in that the experiment apparatus contained the typical ratio of fluid to typical transformer internal materials like conductor and cellulose. Table E.1 and Table E.2 compare mineral oil and soybean-based ester insulating liquid under similar experimental conditions fully reviewed in DGA Interpretation in Natural Ester Fluids [B7].

The laboratory-based experiment used a test setup including a stainless steel vessel with a capacity of 1.8 liter and an aluminum top sealed with Viton O-rings. The vessel was fitted with inlet and outlet ports and valves using a sample port with septum for liquid sampling. The test cell internal components contained 1.0 liter (approx) of unused natural ester 900 g by weight or 870 g of mineral oil. The copper conductor was wrapped with thermally upgraded paper with a calculated 21 grams of paper. The conductor had 14 turns in two layers wrapped around a 3.5 inch diameter block, which is in the same proportions as used in 16,400 liters scaled down to 1.2 L of oil. The liquid was purged with nitrogen to remove dissolved gases. The liquid was circulated over the hot spot area at a rate slow enough to prevent the temperature to drop significantly.

The liquid was heated to the preselected temperatures for a set time interval. At the end of the heating time, an oil sample was removed with a syringe and analyzed for DGA. Each oil was heated separately to the following hot spots 200 °C for five days; 300 °C for one day; and 500 °C for twenty minutes.

E.1.1 Comparison of mineral oil and soybean- based insulating liquid results

Table E.1 provides a comparison of mineral oil and soybean-based insulating liquids from laboratory hot spot pyrolysis experiments where the hot spot does not include nearby cellulose or paper. The experiments employed scaled-down materials normally found in an electrical transformer for the laboratory experiment environment.

**Table E.1— Laboratory experiment for pyrolysis of soybean-based ester liquid
[μL/L (ppm)]**

EPRI pyrolysis data	200 °C		300 °C		500 °C	
	Mineral oil	Soybean	Mineral oil	Soybean	Mineral oil	Soybean
Hydrogen (H ₂)	4	3	23	460	900	1,300
Methane (CH ₄)	560	43	220	770	6,800	7,900
Ethane (C ₂ H ₆)	290	1,400	170	1,700	9,000	3,000
Ethylene (C ₂ H ₄)	30	35	220	770	14,000	62,000
Acetylene (C ₂ H ₂)	0	0	0	0	0	20
Carbon Monoxide (CO)	280	200	420	4,300	140	11,000
Carbon Dioxide (CO ₂)	2,400	6,200	25,000	17,000	3,000	77,000

E.2 Arcing laboratory experiment summary [B7]

The bench-top arcing device was designed to mimic arcing between contacts of a load tap changer. The custom-made metal container (18.9 L/5 gallons) was fitted with valves, sample ports, purge ports, and electrical feeds that allow arcing between two elkonite contacts submerged in the liquids. The arcing rate was adjusted with an air actuator and a commercial welding unit power supply.

The equipment was filled with test insulating liquid and then sealed. The switch rate was set to 50 per minute for a total of 5,000 transits at a current setting of 140 amps. The liquid temperature was reported at 20 °C to 24 °C and the head space purged with nitrogen prior to test.

Table E.2—EPRI arcing result data [$\mu\text{L/L}$ (ppm)]

	Mineral Oil	Soybean Fluid
Hydrogen (H_2)	1,430	380
Methane (CH_4)	159	35
Ethane (C_2H_6)	40	15
Ethylene (C_2H_4)	572	410
Acetylene (C_2H_2)	3,510	1,440
Carbon Monoxide (CO)	0	80

Annex F

(informative)

Theory⁸

Ester liquids used in transformers are natural esters of glycerol and fatty acids (Figure F.1 and Figure F.2) or synthetic esters of pentaerythritol and organic acids (Figure F.3). The abundance of the polyol backbones, ester groups, and carbon chains of sixteen to eighteen carbons combined with the lack of naphthenic and aromatic structures give ester liquids a notably different composition compared to transformer mineral oils (Figure F.4). Although many of the gases generated under thermal and electrical stress are the same for mineral oils and ester insulating liquids, there are differences in the production, proportions, and solubility of these gases.

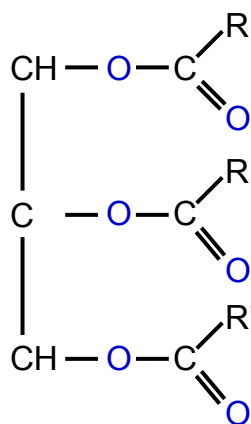


Figure F.1—A natural ester of glycerol and fatty acids



Figure F.2—Common fatty acids found in natural esters

⁸ Reprinted with permission from TJIH2b Analytical Services, Inc., Understanding Dissolved Gas Analysis of Ester fluids-Part 2 Thermal Decomposition of Ester Fluids, ©2012.

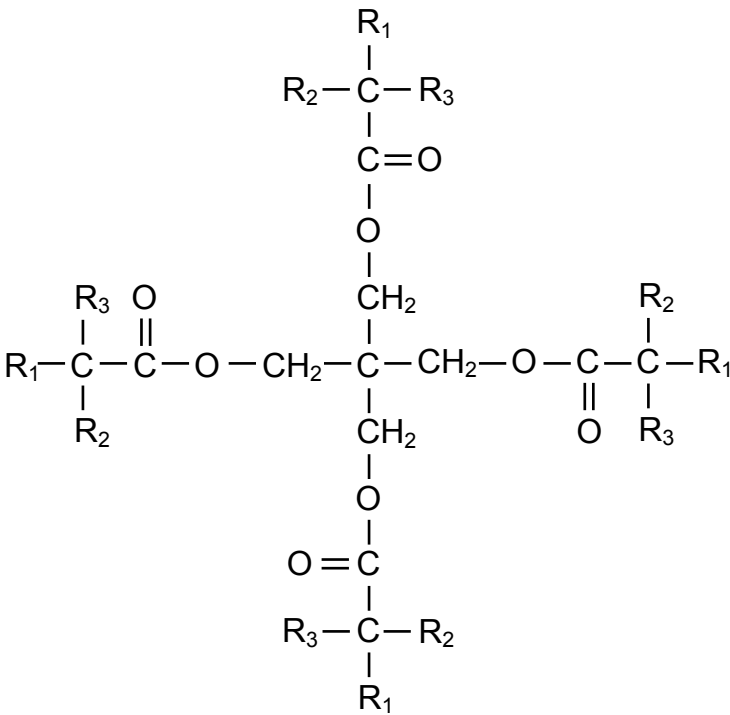


Figure F.3—A synthetic ester pentaerythritol and saturated organic acids

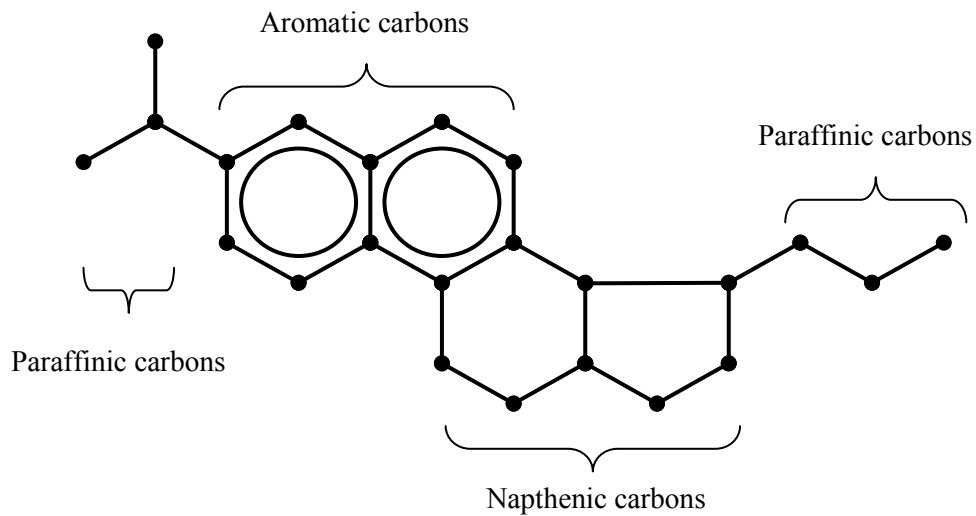


Figure F.4—Common structural features of mineral oil

F.1 Causes of gas formation

The two principal causes of gas formation within an operating transformer are thermal and electrical stresses. Heating due to conditions such as overload, excessive conductor resistance, and magnetic flux leakage can produce gases from thermal decomposition of the liquid and the solid insulation. Gases are also produced from the decomposition of liquid and insulation exposed to electric discharge. Generally, where decomposition gases are formed principally by low energy and partial discharges, there is little or no associated heat. Progressively higher energy discharges will also produce progressively greater amounts of heat energy.

Annex C and Hanson [B11] indicate soybean-based natural ester insulating liquids are susceptible to stray gassing due to interaction with ultraviolet light from sunlight and fluorescent lighting. The sample containers must be stored in dark cool locations and returned to the laboratory for analysis as soon as possible after drawing the sample. This is considered a good practice for all DGA samples.

F.1.1 Stray gassing

Unsaturated fatty acid esters can form hydroperoxides as one of their oxidation products. These hydroperoxides can further react cleaving portions of the fatty acids from the ester molecule. Linolenic fatty acids form hydroperoxides that further react with modest thermal energy to produce ethane (Figure F.5 and Figure F.6). While most natural ester-based dielectric liquids contain a percentage of linolenic acid, the soybean-based liquids contain a higher amount (~7%) and more readily form stray gasses under normal operating conditions.

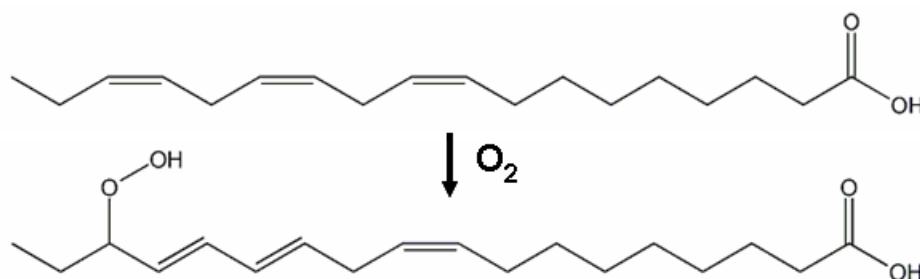


Figure F.5—Oxidation of linolenic acid to 16-hydroperoxy-9Z,12,14E-octadecatrienoic acid

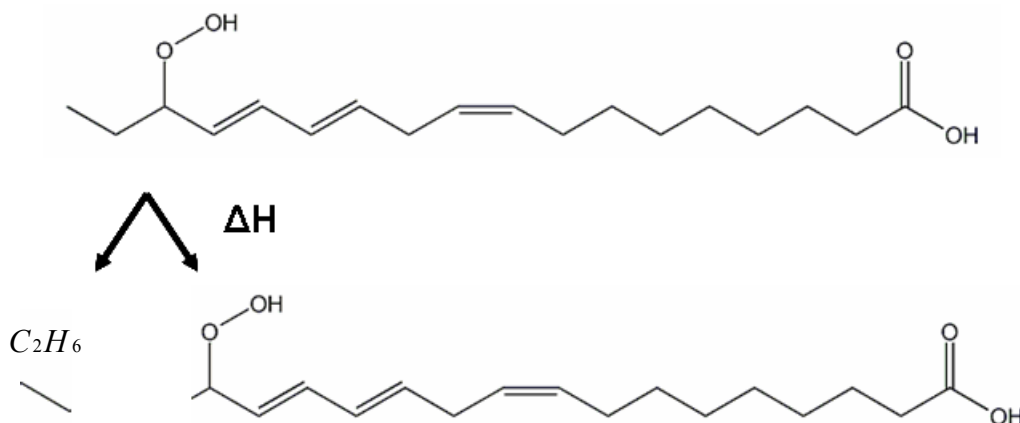


Figure F.6—Formation of ethane from 16-hydroperoxy-9Z,12,14E-octadecatrienoic acid

Ethane from this reaction is produced at normal operating temperatures. Production rates of ethane are increased with temperature. Total ethane production is affected by exposure to oxygen, light, and heat (see Annex C and Annex D).

F.1.2 Liquid thermal decomposition

Sufficient heating of ester liquid produces gases from the thermal decomposition of the hydrocarbon portions of the esters, gases from the thermal decomposition of the ester and acid groups, and gas from the decomposition of an oxidation product of particular group of fatty acids. See Annex D.

F.1.2.1 Acid production

One of the principal ester liquid pyrolysis reactions is the decomposition of the triglyceride to two free fatty acids, propenal (acrolein), and the ketene from the third fatty acid. See Figure F.7. The data in Table D.1, Table D.2, Table D.3, and Table D.4 show increases and/or very high acid content through much of the temperature range of the studies depending on the fatty acid composition of the liquid. By comparison, the data in Table D.5 shows no similar effect for mineral oil. The total acid number data as well as gas chromatography mass spectrometry (GCMS) data confirms this pathway. Specifically, GCMS analysis reveals the presence of the most abundant fatty acids, ketenes, and propenal.

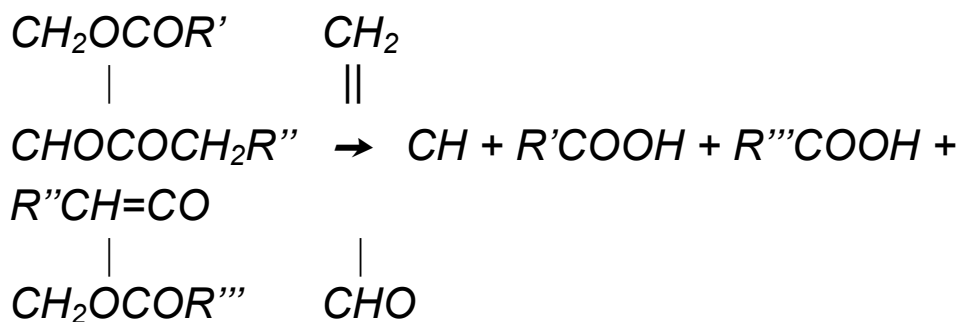


Figure F.7—Thermal decomposition of triglycerides to produce free fatty acids, a ketene, and propenal

While this reaction produces no gaseous products, each of the products reacts further to produce gases.

F.1.2.2 Carbon oxides and ethylene production

Fundamental research associated with the reaction shown in Figure F.7, indicates that the products formed by that process are further reacted according to the reactions shown in Figure F.8 through Figure F.12.

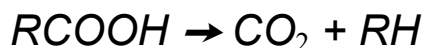


Figure F.8—Thermal decomposition of fatty acids to produce carbon dioxide



Figure F.9—Thermal decomposition of fatty acids to produce carbon monoxide



Figure F.10—Thermal decomposition of fatty acids to produce carbon dioxide, carbon monoxide, and ethylene



Figure F.11—Thermal decomposition of ketenes to produce carbon monoxide

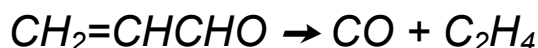


Figure F.12—Thermal decomposition of propenal to produce carbon monoxide and ethylene

Examination of the data in Table D.1 through Table D.4 shows large increases in carbon oxides that correspond to increases in acid content. By comparison, the carbon oxide levels from mineral oil in Table D.5 indicate the significance of these increases. As with the triglyceride decomposition reaction, DGA and GCMS analysis confirms the activity of these pathways. Examination of Figure F.8 through Figure F.12 shows that the greatest number of reaction pathways produce carbon monoxide. This offers the possibility of greater production of carbon monoxide compared to carbon dioxide. The experimental data in Table D.1 through Table D.4, clearly demonstrates that carbon monoxide is more abundant than carbon dioxide.

Examination of Figure F.8 through Figure F.12 shows that a significant amount of ethylene is produced while carbon monoxide is being produced. Because ethylene production in mineral oil at the highest temperatures is very similar to that in the natural ester liquids, it is more difficult to casually observe the contributions to ethylene from these reactions (Figure F.10 and Figure F.12). The contributions at 450 °C, 500 °C, and 550 °C may provide the most obvious indications.

F.1.2.3 Hydrogen and hydrocarbon production

The hydrocarbon portions of the fatty acid components are more similar to the carbon structures of mineral oil and are subject to similar reactions. These reactions are shown in Figure F.13 through Figure F.17.



Figure F.13—Decomposition into elements that produces hydrogen



Figure F.14—Dehydrogenation of paraffins to olefins produces hydrogen and can produce ethylene



Figure F.15—Splitting decomposition of paraffins can produce methane, ethane, and ethylene



Figure F.16—Aromatic cyclization of paraffins produces hydrogen



Figure F.17—Dehydrogenation of olefins to acetylenic hydrocarbons produces hydrogen and can produce acetylene

There are three noteworthy observations about these pathways. First, hydrogen is produced by four of these five reactions; yet the abundance of hydrogen is less than any one of the hydrocarbon gases. This suggests that the hydrogen producing reactions are considerably less active than the hydrocarbon producing reactions or considerable amounts of hydrogen are being consumed in other reactions. Second, two of these reactions (Figure F.14 and Figure F.15) can produce ethylene. Ethylene that is produced by the reactions in Figure F.10 and Figure F.12 could never exceed the amount of carbon monoxide. Comparison of ethylene to carbon monoxide levels indicates that one or both of the reactions shown in Figure F.14 and Figure F.15 must become very active at higher temperatures.

Third, only one of these reactions can produce methane and ethane. Interestingly, through part of the higher temperature range methane and ethane are produced in similar quantities for natural ester liquids. In mineral oils, this is not the case. The short hydrocarbon sequences of mineral oils and the long sequences of natural ester liquids may be part of the explanation for this difference.

Annex G

(informative)

Duval Triangle analysis method

The Duval Triangle method (see Figure G.1) can be applied when DGA results are above minimum gas concentrations and there is a suspected fault present in the equipment. The Duval Triangle regions are based on the increasing level of experience with natural and synthetic ester liquids. Regions are established through empirical inspection of DGA results from a specific liquid type and the observed equipment gassing source. The regions are updated as new data and information become available over time. Methane (CH₄), ethylene (C₂H₄), and acetylene (C₂H₂) are evaluated using the ratios found in Table G.1 with each ratio term plotted on the triangle illustration found in the example in G.1.

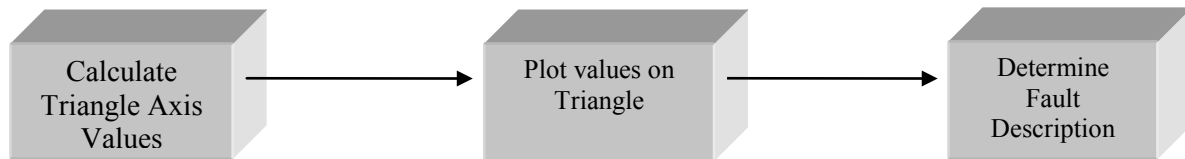


Figure G.1—Duval Triangle method

Table G.1—Duval Triangle formula

Triangle axis	Formula
%CH ₄	$= \text{CH}_4 / (\text{CH}_4 + \text{C}_2\text{H}_4 + \text{C}_2\text{H}_2)$
%C ₂ H ₄	$= \text{C}_2\text{H}_4 / (\text{CH}_4 + \text{C}_2\text{H}_4 + \text{C}_2\text{H}_2)$
%C ₂ H ₂	$= \text{C}_2\text{H}_2 / (\text{CH}_4 + \text{C}_2\text{H}_4 + \text{C}_2\text{H}_2)$

G.1 Example

The example parameters are calculated from DGA data based on Table G.1. The example parameters are plotted on the Duval Triangle as noted by the dotted plot lines. The D2 area symbol is matched with the fault description code found in Table G.2, which indicates possible high energy discharges as the gassing source.

Duval Triangle images based on the most recent understanding of the liquid behaviors are presented in Figure G.2, Figure G.3, and Figure G.4.

Table G.2—Duval Triangle example parameters

Example parameters	
Triangle axis	
%CH ₄	10
%C ₂ H ₄	50
%C ₂ H ₂	40

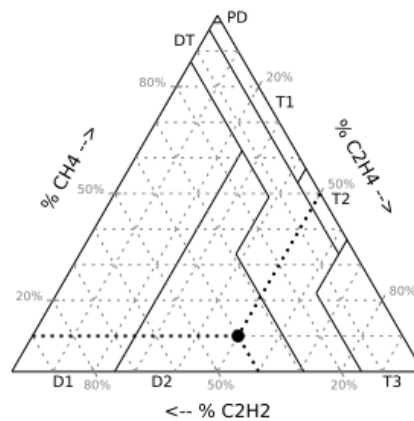


Figure G.2—Duval Triangle with plotted example parameters

Table G.3—Duval fault type symbols with highlighted example fault description

Area symbol	Fault description
PD	Partial discharges
D1	Discharges of low energy
D2	Discharges of high energy
DT	Thermal fault or electrical discharge
T1	Thermal fault, $T < 300\text{ }^{\circ}\text{C}$
T2	Thermal fault, $300^{\circ}\text{C} < T < 700\text{ }^{\circ}\text{C}$
T3	Thermal fault, $T > 700\text{ }^{\circ}\text{C}$

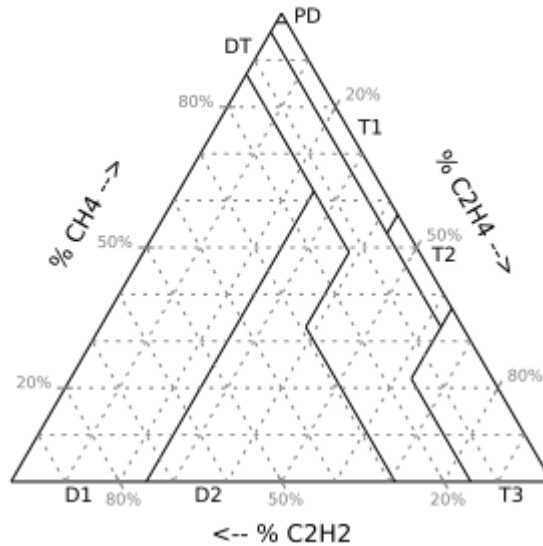


Figure G.3—Soybean-based ester liquid Duval Triangle

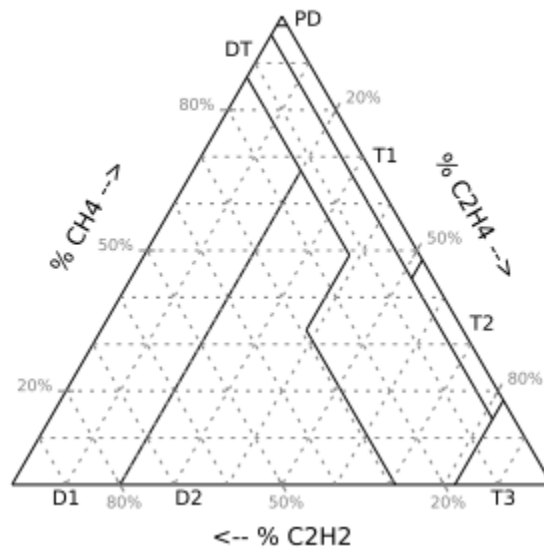


Figure G.4—High oleic sunflower-based ester liquid Duval Triangle

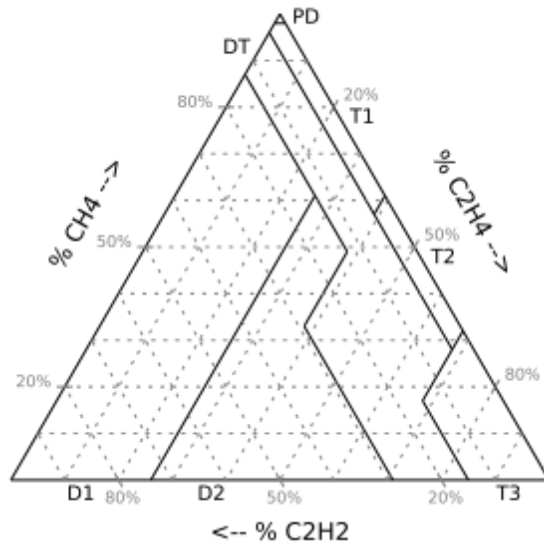


Figure G.5—Synthetic ester liquid Duval Triangle

Annex H

(informative)

Case study

H.1 Soybean-based natural ester liquid case study⁹ [B5]

A 230 kV, 8 MVA substation transformer installation using soybean-based insulating liquid that was placed into service as a bank of three single phase transformers with one on-site spare transformer. Several samples were taken immediately after the transformer was energized to establish a baseline for trending and to monitor it for proper operation. Sample 1 reported a low amount of acetylene, which raised concerns about a developing problem. The sample data is shown in the Table H.1.

Table H.1—Case study DGA result data [μL/L (ppm)]

Sample Data (ppm)	1	2	3	4	5
Sample Date	28 DEC 2005	18 JAN 2006	3 AUG 2006	31 JUL 2007	16 AUG 2007
Days between Sample	—	20	195	358	16
Hydrogen (H ₂)	6	5	16	280	268
Oxygen (O ₂)	3,244	1,735	545	712	1,539
Nitrogen (N ₂)	50,085	62,708	60,086	42,571	50,300
Methane (CH ₄)	0	1	2	55	60
Carbon Monoxide (CO)	11	10	31	86	91
Carbon Dioxide (CO ₂)	309	576	937	756	1,582
Ethylene (C ₂ H ₄)	2	2	3	46	49
Ethane (C ₂ H ₆)	0	2	19	31	35
Acetylene (C ₂ H ₂)	2	2	2	226	217

Table H.2—Case study DGA rate of change (ppm/day)

Sample Date	18 JAN 2006	3 AUG 2006	31 JUL 2007	16 AUG 2007
Hydrogen (H ₂)	−0.1	0.1	0.7	−0.8
Oxygen (O ₂)	−75.5	−6.1	0.5	51.7
Nitrogen (N ₂)	631.2	−13.4	−48.9	483.1
Methane (CH ₄)	0.1	0.0	0.1	0.3
Carbon Monoxide (CO)	−0.1	0.1	0.2	0.3
Carbon Dioxide (CO ₂)	13.4	1.9	−0.5	51.6
Ethylene (C ₂ H ₄)	0.0	0.0	0.1	0.2
Ethane (C ₂ H ₆)	0.1	0.1	0.0	0.3
Acetylene (C ₂ H ₂)	0.0	0.0	0.6	−0.6

⁹ Reprinted with permission from Hartford Steam Boiler Inspection and Insurance Co., TOGA[®] Presentation: Review of Case Studies, ©2013.

The subsequent samples #3, #4, and #5 reported similar DGA gas profiles from the Duval Triangle method. There was an increase in the individual gas concentration rates of changes for these samples. Sample #4 indicated arcing activity so the confirmation sample #5 was quickly taken. The transformer owner made the decision to remove the transformer for repair and installed the spare transformer.

The combustible gas concentrations rates of changes in Table H.2 show an upward trend in many of the gases. There are slight increases in the carbon monoxide and carbon dioxide gas concentrations that can be related to cellulose material (paper insulation) aging or overheating from the gassing source. The analysis included use of the Duval Triangle method. The Duval Triangle method could not be used on the earlier three samples because of the low combustible gas concentrations. The gassing source was high current amperage through the shield ground conductor. The ground conductor was routed near thin cellulose wrapping that burned along the conductor run. The loose shield ground connection would flashover from the increasing shield electrical potential to produce increases in the acetylene gas concentration. The subsequent internal inspection found a loose ground connection on a shield.

Table H.3—Case study calculated Duval Triangle DGA method ratios

Ratios	31-Jul-07	16-Aug-07
% CH ₄	16.8%	18.4%
%C ₂ H ₄	14.1%	15.0%
%C ₂ H ₂	69.1%	66.6%
Diagnostic code	Discharges of low energy D1	Discharges of low energy D1

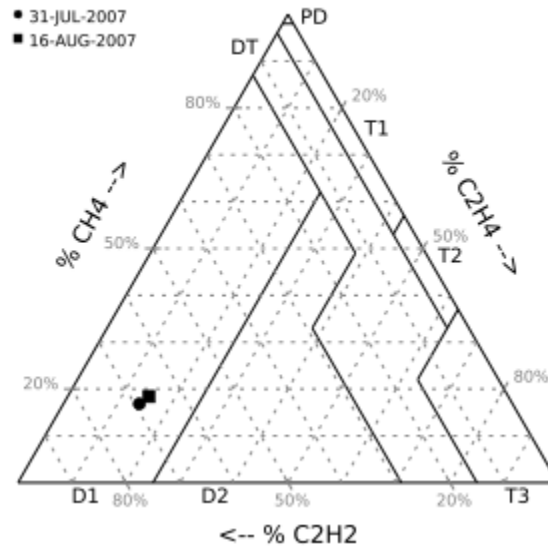


Figure H.1—Soybean-based ester insulating liquid Duval Triangle with case study results plots

Another aspect of this case study is the evolution of ethane during normal operation. The four transformers with the original charge of soybean-based insulating liquid did not show ethane production. Following the T3 transformer repair and liquid replacement, ethane was reported in subsequent DGA samples. The bank of transformers was placed into service at the end of 2005. The T1 transformer was removed from service in 2007, repaired and refilled with new liquid (T1 post repair data set). Post repair samples on T1 reported increasing ethane gas concentrations that appeared to stabilize over time (Table H.4).

Table H.4—Examples of ethane evolution from case study result data [$\mu\text{L/L}$ (ppm)]

Unit	Sample date	Hydrogen	Oxygen	Nitrogen	Methane	Carbon monoxide	Carbon dioxide	Ethylene	Ethane	Acetylene
T1 pre-repair	18-Jan-06	6	533	61,792	0	10	538	1	1	2
T2	31-Jul-07	44	756	41,958	2	18	599	1	17	1
T3 spare	31-Jul-07	38	3,901	47,000	2	26	653	2	44	1
T4	24-Oct-07	27	3,353	55,951	1	28	588	0	0	0
T1 post repair	14-Sep-10	2	387	62,566	2	43	477	3	231	0

H.2 Synthetic ester insulating liquid case studies

The two transformers described below showed high levels of hydrogen and methane in service. The transformers were dismantled and visually inspected. The root cause analysis confirmed partial discharge in the windings. Deposits of X-wax were also found. The affected windings were replaced and the transformers are again in service without excessive gas generation.

Case Study for Synthetic Insulating Liquid

The sealed distribution transformer was rated at 20 kV with a capacity of 3.5 MVA and it was manufactured in 2008. The insulation system consisted of flame resistant synthetic fibers solid insulation and a synthetic insulating liquid.

Table H.5—DGA result data for synthetic-based ester insulating fluid case study [$\mu\text{L/L}$ (ppm)]

Sample data	Sample date	Days between sample	Hydrogen	Methane	Carbon monoxide	Carbon dioxide	Ethylene	Ethane	Acetylene
1	27-Apr-10		1,281	319	74	1,332	1	32	0
2	9-Jun-10	32	1,456	161	66	846	1	16	0

Table H.6—Synthetic-based ester insulating fluid case study Duval Triangle ratio analysis

Sample data	Sample date	% CH_4	% C_2H_4	% C_2H_2	Diagnostic code
1	27-Apr-10	99.7%	0.3%	0.0%	PD—Partial discharges
2	9-Jun-10	99.4%	0.6%	0.0%	PD—Partial discharges

Annex I

(informative)

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